

Problem Set 4

This homework must be turned in on Brightspace by December 6th 2022, 11:59pm (Paris time). It must be your own work, and your own work only you must not copy anyones work, or allow anyone to copy yours. This extends to writing code. You may consult with others, but when you write up, you must do so alone. Your homework submission must be written and submitted using Jupyter Notebook. No handwritten solutions will be accepted. You should submit:

- A compiled PDF file named `yourNetID_solutions.pdf` containing your solutions to the problems.
- A `.ipynb` file containing the code and text used to produce your compiled pdf named `yourNetID_solutions.ipynb`. Note that math can be typeset in markdown cells (in the jupyter notebook) in the same way as LaTeX.

Please make sure your answers are clearly structured in the jupyter notebook file:

- Label each question (e.g. 1.1 for exercise 1, question 1).
- Do not include written answers as code comments.
- The code used to obtain the answer for each question part should accompany the written answer.

Exercise 1 About difference in differences (Total: 25 points)

Suppose you have a time-series dataset with repeated observations of the same n units over two time periods. No unit is treated at $t = 0$, and some units are treated at $t = 1$, each unit's treatment status at $t = 1$ is indicated by the random variable T_i . Suppose that exactly n_t units are treated, and n_c units are in the control group, so that $P(T_i = 1) = \frac{n_t}{n}$. You know that observed outcomes take the following form:

$$\begin{aligned} Y_{i0} &= \delta_0 + u_i + \epsilon_{i0}, \\ Y_{i1} &= \tau_i T_i + \delta_1 + u_i + \epsilon_{i1}, \end{aligned}$$

where δ_ζ , u_i , and τ_i are constants, and T_i and $\epsilon_{i\zeta}$ are random variables that satisfy $E[\epsilon_{i1}|T_i = t] = E[\epsilon_{i0}|T_i = t] = \eta_i(t)$ for some constants $\eta_i(t)$ for both $t = 1$ and $t = 0$, and that $\eta_i(1) \neq \eta_i(0)$. Note that t represents the treatment, and ζ the time, as in the lecture.

Question 1 (5 points)

In this setting, the time-dependent potential outcome under control is defined as:

$$Y_{i\zeta}(0) = \delta_\zeta + u_i + \epsilon_{i\zeta}$$

for $\zeta = 0, 1$. Show that the parallel trends assumption is satisfied in this setting by showing that $E[Y_{i1}(0) - Y_{i0}(0)|T_i = 1] = E[Y_{i1}(0) - Y_{i0}(0)|T_i = 0]$.

Question 2 (10 points)

Show that the estimator:

$$\hat{\tau} = \frac{1}{n_t} \sum_{i=1}^n (Y_{i1} - Y_{i0})T_i - \frac{1}{n_c} \sum_{i=1}^n (Y_{i1} - Y_{i0})(1 - T_i)$$

is unbiased for the ATE, i.e., show that $E[\hat{\tau}] = \frac{1}{n} \sum_{i=1}^n \tau_i$ in this setting.

Question 3 (10 points)

Suppose that we ignore the first time period, and we instead use the Neyman estimator with only the data for $\zeta = 1$ for the ATE, that is, we use:

$$\hat{\tau}_{\zeta=1} = \frac{1}{n_t} \sum_{i=1}^n Y_{i1}T_i - \frac{1}{n_c} \sum_{i=1}^n Y_{i1}(1 - T_i),$$

derive a formula for the bias of $\hat{\tau}_{\zeta=1}$ in terms of the constants $\eta_i(t)$; that is, derive a formula for the quantity: $E[\hat{\tau}_{\zeta=1}] - \frac{1}{n} \sum_{i=1}^n \tau_i$.

Exercise 2 Instrumental variables (40 points)

Despite heated political and media rhetoric, there are few causal estimates of the effect of expanded healthcare insurance on healthcare outcomes. One landmark study, the [Oregon Health Insurance Experiment], covered new ground by utilizing a randomized control trial implemented by the state government of Oregon. To allocate a limited number of eligible coverage slots for the states Medicaid expansion, about 30,000 low-income, uninsured adults (out of about 90,000 wait-list applicants) were randomly selected by lottery to be allowed to apply for Medicaid coverage. Researchers collected observable measures of health (blood pressure, cholesterol, and blood sugar levels), as well as hospital visitation and healthcare expenses for 6,387 selected adults and 5,842 not selected adults.

For this problem, you will need the **OHIE.dta** file. The variables you will need are:

treatment - Selected in the lottery

ohp_all_ever_admin - Ever enrolled in Medicaid from matched notification date to September 30, 2009 (actually took the treatment)

tab2bp_hyper - Outcome: Binary indicator for elevated blood pressure (defined a systolic pressure of 140mm Hg or more and a diastolic pressure of 90mm Hg or more)

tab2phqtot_high - Outcome: Binary indicator for a positive screening result for depression (defined as a score of 10 or higher on the Patient Health Questionnaire - 8)

tab4_catastrophic_exp_inp - Outcome: Indicator for catastrophic medical expenditure (total out-of-pocket medical expenses $\geq$ 30% of household income)

tab5_needmet_med_inp - Outcome: Participant feels that they received all needed medical care in past 12 months (binary indicator)

Question 1 Data preparation (5 points)

There are a few preliminary steps to ensure the dataset is ready to use.

1. Restrict the dataset to the needed variables.
2. Convert all the variables to numerical values if needed.
3. Remove the observations with missing values.
4. Can this last step induce bias? Which type of bias? Briefly explain. For the rest of the problem, we will neglect this issue.

Question 2 (10 points)

Estimate the intent-to-treat effects of assignment to treatment (being eligible to apply) on each of the four outcomes (elevated blood pressure, depression, catastrophic medical expenditure, and whether respondents had their health care needs met) with the difference-in-means estimator. Provide 95% confidence intervals for each estimate and interpret your results.

Question 3 (10 points)

Suppose that researchers actually wanted to estimate the effect of Medicaid enrollment on each of the four outcomes. Suppose they first used the naive difference-in-means estimator with the indicator of Medicaid enrollment as a treatment. Report a 95% confidence interval for each of your estimates and interpret your results. Why might these be biased estimates for the causal effect of Medicaid enrollment?

Question 4 (5 points)

Suppose we were to use assignment to treatment as an instrument for actually receiving Medicaid coverage.

Consider that not everyone who was selected to apply for Medicaid actually ended up applying and receiving coverage. Likewise, some applicants who were not selected to receive the treatment nevertheless were eventually covered. What were the compliance rates (the level of Medicaid enrollment) for subjects who were selected and subjects who were not selected? Report the rate of Medicaid enrollment for the treated and the untreated, as well as the correlation between the treatment and Medicaid enrollment. Is the instrument of assignment-to-treatment a strong instrument for actual Medicaid enrollment? What other assumptions do you need to identify the LATE in the subpopulation of compliers? Are they enforced here?

Question 5 (10 points)

Now estimate the LATE of Medicaid enrollment on each of the four outcomes using an instrumental variables strategy (you can use the `IV2SLS` function from the module `linearmodels.iv`). Report a 95% confidence interval for your estimates and interpret your results. Compare the estimates to those you obtained in Question 3.

Question 6 (Bonus question - 5 points)

What additional assumptions do you have to make in order to interpret your estimates from Question 5 as an Average Treatment Effect for the entire sample?